

1번) 다음 그림은 LCD를 이용한 회로도이다.

다음과 같이 동작하도록 tinkercad에 회로+코드를 완성하라. (100점)

```
#include <Wire.h>
#define TEMP_ADDR 72

//Include the LCD library and initialize:
#include <LiquidCrystal.h>
LiquidCrystal lcd(5,6,7,8,9,10);

//Custom degree character
byte degree[8] = {
  B00110,
  B01001,
  B01001,
  B00110,
  B00000,
  B00000,
  B00000,
  B00000,
};

//Custom "Fan On" indicator
byte fan_on[8] = {
  B00100,
  B10101,
  B01110,
  B11111,
  B01110,
  B10101,
  B00100,
  B00000,
};

//Custom "Fan Off" indicator
byte fan_off[8] = {
  B00100,
  B00100,
  B00100,
  B11111,
  B00100,
  B00100,
  B00100,
  B00000,
};
```

```

//Pin Connections
const int RED      =2;
const int BLUE     =3;
const int GREEN    =4;
const int DOWN_BUTTON =11;
const int UP_BUTTON  =12;
const int FAN       =13;
const int temp = 0;

//Variables for debouncing
boolean lastDownTempButton = LOW;
boolean currentDownTempButton = LOW;
boolean lastUpTempButton = LOW;
boolean currentUpTempButton = LOW;

int set_temp = 25;    //The Default desired temperature
boolean one_time = false; //Used for making the speaker beep only one time

void setup()
{
  pinMode(FAN, OUTPUT);
  pinMode(RED, OUTPUT);
  pinMode(GREEN, OUTPUT);
  pinMode(BLUE, OUTPUT);

  //Turn off the LED
  //It is common-anode, so setting the cathode pins to HIGH turns the LED off
  digitalWrite(RED, LOW);
  digitalWrite(GREEN, LOW);
  digitalWrite(BLUE, LOW);
  //Create a wire object for the temp sensor
  //Wire.begin();

  //Set up the LCD's number of columns and rows
  lcd.begin(16, 2);
  Serial.begin(9600);
  //Make custom characters
  lcd.createChar(0, degree);
  lcd.createChar(1, fan_off);
  lcd.createChar(2, fan_on);

  //Print a static message to the LCD
  //Print a static message to the LCD
  lcd.setCursor(0,0);
  lcd.print("Current:");
  lcd.setCursor(10,0);
  lcd.write((byte)0);

```

```

lcd.setCursor(11,0);
lcd.print("C");
lcd.setCursor(0,1);
lcd.print("Set:");
lcd.setCursor(10,1);
lcd.write((byte)0);
lcd.setCursor(11,1);
lcd.print("C");
lcd.setCursor(15,1);
lcd.write(1);
}

```

//A debouncing function that can be used by multiple buttons
boolean debounce(boolean last, int pin)

```

{
    boolean current = digitalRead(pin);
    if (last != current)
    {
        delay(5);
        current = digitalRead(pin);
    }
    return current;
}

```

void loop()

```

{
    //Get the Temperature
    // Wire.beginTransmission(TEMP_ADDR); //Start talking
    // Wire.write(0);                      //Ask for register zero
    // Wire.endTransmission();             //Complete transmission
    // Wire.requestFrom(TEMP_ADDR, 1);    //Request 1 byte
    // while(Wire.available() == 0);      //Wait for response
    // int c = Wire.read();                 //Get the temp in C
    int c = analogRead(temp);
    Serial.println(c);
    c = round(0.48*c - 49.64);
    lcd.setCursor(8,0);                    //Move the cursor
    lcd.print(c);                          //Print this new value

    //Debounce both buttons
    currentDownTempButton = debounce(lastDownTempButton, DOWN_BUTTON);
    currentUpTempButton = debounce(lastUpTempButton, UP_BUTTON);

    //Turn down the set temp
    if (lastDownTempButton== LOW && currentDownTempButton == HIGH)
    {
        set_temp--;
    }
}

```

```

}
//Turn up the set temp
else if (lastUpTempButton== LOW && currentUpTempButton == HIGH)
{
    set_temp++;
}
//Print the set temp
lcd.setCursor(8,1);
lcd.print(set_temp);
lastDownTempButton = currentDownTempButton;
lastUpTempButton = currentUpTempButton;

//It's too hot!
if (c >= set_temp)
{
    lcd.setCursor(15,1);
    lcd.print(" ");
    digitalWrite(FAN, HIGH);
    lcd.setCursor(15,0);
    lcd.write(2);
    digitalWrite(RED, HIGH);
    digitalWrite(GREEN, LOW);
    digitalWrite(BLUE, HIGH);
}
//It's not too hot!
else
{
    lcd.setCursor(15,0);
    lcd.print(" ");
    digitalWrite(FAN, LOW);
    lcd.setCursor(15,1);
    lcd.write(1);
    digitalWrite(RED, HIGH);
    digitalWrite(GREEN, HIGH);
    digitalWrite(BLUE, LOW);
}
}

```

2번) 다음중 I²C 통신에 관한 설명중 맞는 것을 모두 고르시오. (15점)

c, d, h

3번) SPI 통신에서 아두이노에 센서장치를 5개를 연결할 경우 필요한 연결선을 몇개인가? (5점)

8개

4번) 다음중 Bluetooth 기술과 관련이 있는 것을 모두 고르시오.

b, d, g

5번) 스마트폰 Bluetooth와 아두이노 bluetooth를 이용하여 전등제어와 동시에 스피커에서 멜로디를 발생시키려고 한다. 교재 코드 16-3의 loop() 내부를 수정해서 melody가 나오도록 하려고 한다. 아래의 코드의 빈칸을 채워라. (30점)

```
for (int i = 0; i < 3; i++)
{
    tone(SPEAKER, melody[i], times[i]);
    delay(times[i]);
}
```